

FUNCTION

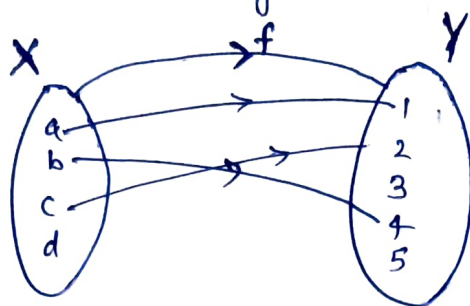
Defⁿ:- Let X and Y are two non-empty sets. Then f is a rule which associates every element of X to a unique element of Y i.e.

$$f: X \rightarrow Y$$

- (i) Existence of Image
- (ii) Uniqueness of Image

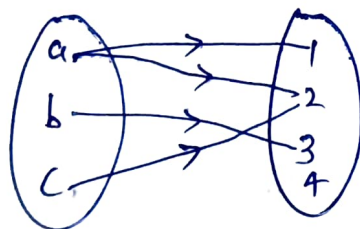
Examples

(a)



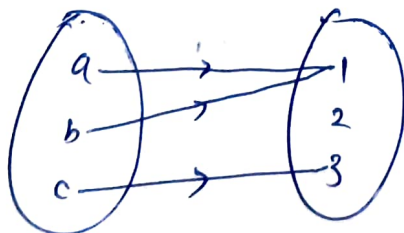
Since $f(d)$ DNE, so it is not a function

(b)



Not a function due to not uniqueness of ~~a function~~ Image

(c)



Yes, It is a function

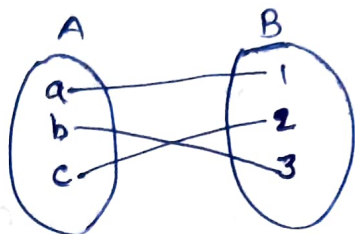
* Domain of f : Elements of X

* Co-Domain of f : Elements of Y

* Range of f : $f(X) := \{f(x) \in Y : x \in X\}$

Types of Function:

- ① One-one Function: A function $f: A \rightarrow B$ is said to be one-one if every element of X has unique and distinct Image in Y .



$$\text{i.e. } |\text{Domain}| = |\text{Range}|$$

$$\text{i.e. } f(x) = f(y) \Rightarrow x = y$$

or

$$\text{If } x \neq y \Rightarrow f(x) \neq f(y)$$

Graphical Method:

If any line parallel to x -Axis cuts the graph more than once. Then It is not one-one function.

Calculus Method:

If f is differentiable and $f'(x) > 0$ or $f'(x) < 0$ and $f'(x) = 0$ only at countable points. Then f is one-one function.

Note:

A function which is not one-one is called many one function.

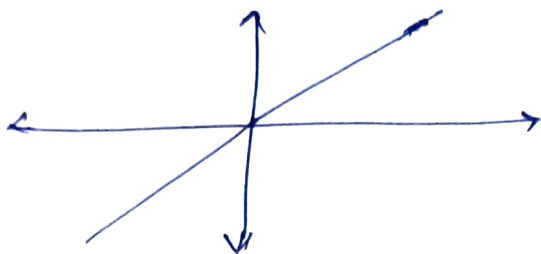
1. Algebraic Functions:

(i) Polynomial Function:

(1) Identity Function:

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = x$$

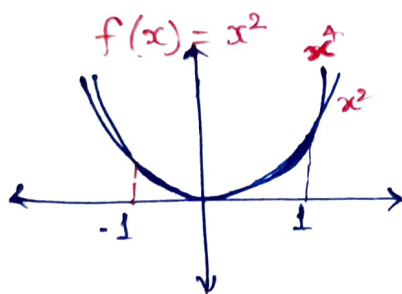


$$D_f = \mathbb{R}$$

$$R_f = \mathbb{R}$$

odd function
one-one function
onto

(2)



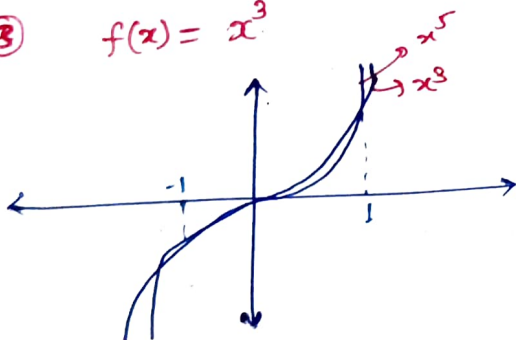
$$D_f = \mathbb{R}$$

$$R_f = [0, \infty)$$

Even function
Not one-one

(3)

$$f(x) = x^3$$



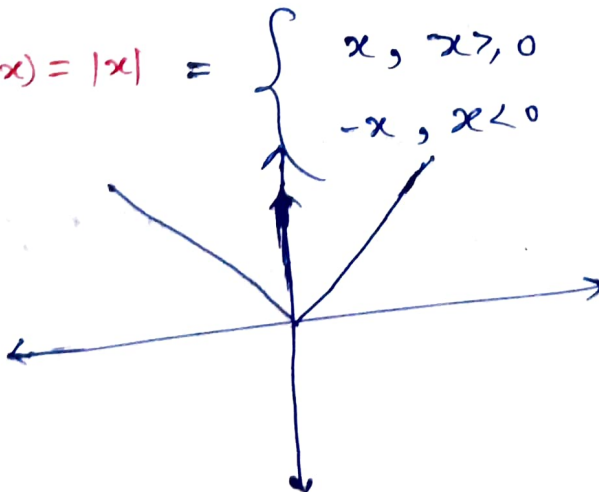
$$D_f = \mathbb{R}$$

$$R_f = \mathbb{R}$$

odd function
one-one function

(4)

$$f(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



$$D_f = \mathbb{R}$$

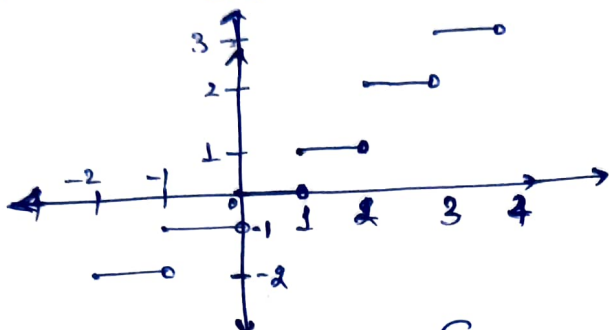
$$R_f = [0, \infty) = \mathbb{R}^+ \cup \{0\}$$

Even function
Not one-one
Not on-to

5. Greatest Integer Function:

$$f(x) = [x] \text{ or } \lfloor x \rfloor$$

$$* \quad x-1 < [x] \leq x$$



$$D_f = \mathbb{R}$$

$$R_f = \mathbb{Z}$$

Neither odd nor even

Not one-one

$$(i) \quad [x] + [-x] = \begin{cases} 0 & ; x \in \mathbb{Z} \\ -1 & ; x \notin \mathbb{Z} \end{cases}$$

For $x \in \mathbb{Z}$; Let $x = -2$

$$[-2] + [2] = 0$$

For $x \notin \mathbb{Z}$; Let $x = 1.6$

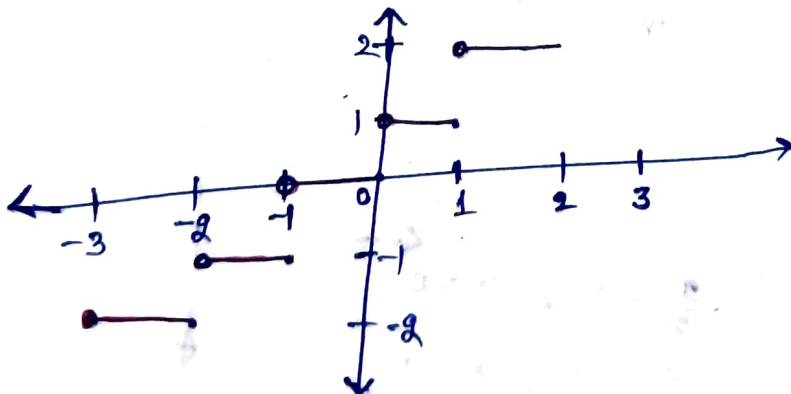
$$[1.6] + [-1.6] = 1 - 2 = -1$$

$$[-2.6] = -3$$

6. Least (ceiling) Integer Function:

$$f(x) = \lceil x \rceil$$

$$x \leq \lceil x \rceil < x+1$$



$$D_f = \mathbb{R}$$

$$R_f = \mathbb{Z}$$

Neither odd
~~neither~~ even

Not one-one
Not onto

$$\lceil -2.6 \rceil = -2$$

7. Fractional part function:

$$f(x) = \{x\} = x - [x]$$

$$\therefore x-1 < [x] \leq x$$

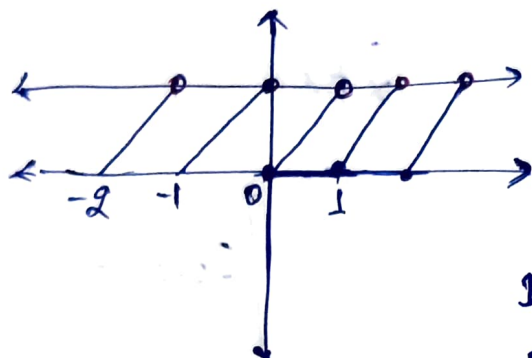
$$1-x > -[x] \geq -x$$

$$1 > x - [x] \geq 0$$

$$0 \leq x - [x] < 1$$

$$0 \leq \{x\} < 1$$

$$\{x\} + \{-x\} = \begin{cases} 0; & x \in \mathbb{Z} \\ 1; & x \notin \mathbb{Z} \end{cases}$$



$$f(x) = \begin{cases} x+2 & ; -2 \leq x \\ x+1 & ; -1 \leq x < 0 \\ x & ; 0 \leq x < 1 \\ x-1 & ; 1 \leq x < 2 \\ x-2 & ; 2 \leq x < 3 \end{cases}$$

$$\mathcal{D}_f = \mathbb{R}$$

$$\mathcal{R}_f = [0, 1)$$

Neither odd nor even

Not one-one

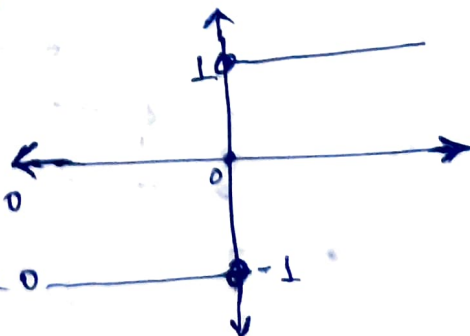
Period = 1

8. Signum Function:-

$$f(x) = \text{sgn}(x)$$

$$f(x) = \begin{cases} \frac{|x|}{x} & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

$$f(x) = \begin{cases} \frac{x}{x} = 1, & x > 0 \\ -\frac{x}{x} = -1, & x < 0 \\ 0 & ; x = 0 \end{cases}$$



$$\mathcal{D}_f = \mathbb{R}$$

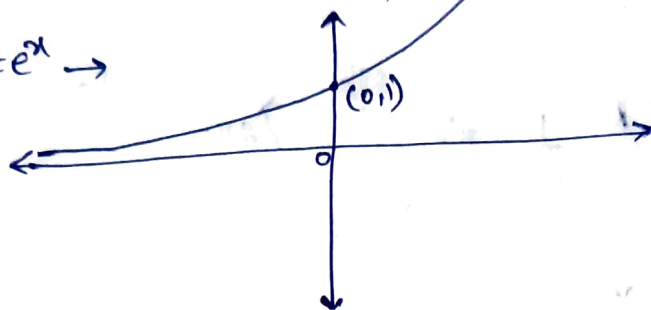
$$\mathcal{R}_f = \{0, -1, 1\}$$

9. Power function:

$$f(x) = a^x$$

when $(a > 1)$

$$f(x) = e^x \rightarrow$$

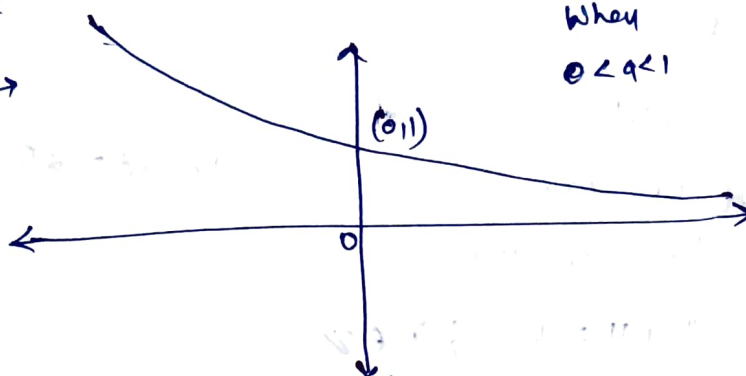


$$D_f = \mathbb{R}$$

$$R_f = (0, \infty)$$

One-one function

$$f(x) = e^{-x} \\ = \left(\frac{1}{e}\right)^x \rightarrow$$



when $0 < a < 1$

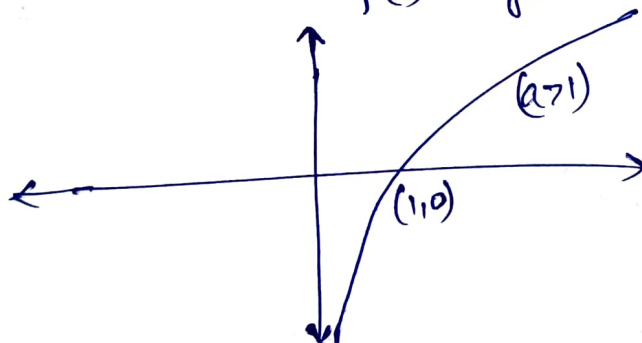
$$D_f = \mathbb{R}$$

$$R_f = (0, \infty)$$

10. Logarithmic function:

$$f(x) = \log_a x$$

($a > 1$)

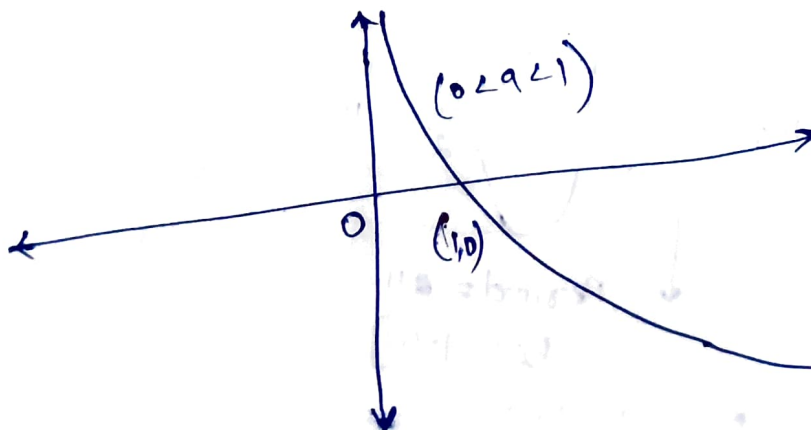


$$D_f = (0, \infty)$$

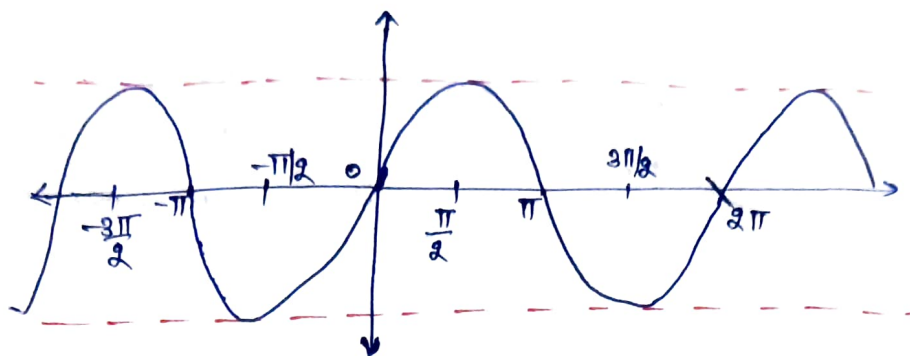
$$R_f = (-\infty, \infty) = \mathbb{R}$$

one-one

($0 < a < 1$)



1). $f(x) = \sin x$



$$D_f = \mathbb{R}$$

$$R_f = [-1, 1]$$

Not one-one

; Period = 2π

odd function

$$\sin n\pi = 0 ; n \in \mathbb{Z}$$

$$\sin\left((2n+1)\frac{\pi}{2}\right) = (-1)^n ; n \in \mathbb{Z}$$

* $\sin x = \sin y$

$$y = \pi - x$$

$$2\pi + x$$

$$3\pi - x$$

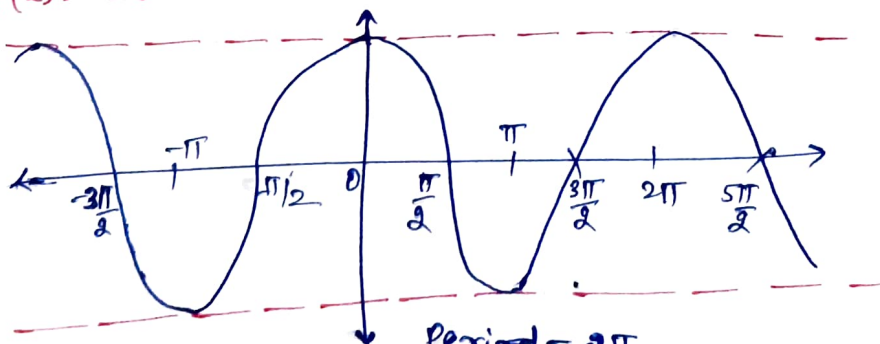
$$4\pi + x$$

$$y = n\pi + (-1)^n x, n \in \mathbb{Z}$$

$$\sin x = \frac{1}{2} = \sin \pi/6$$

$$x = n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$$

12. $f(x) = \cos x$



Period = 2π

$$R_f = [-1, 1]$$

Not one-one

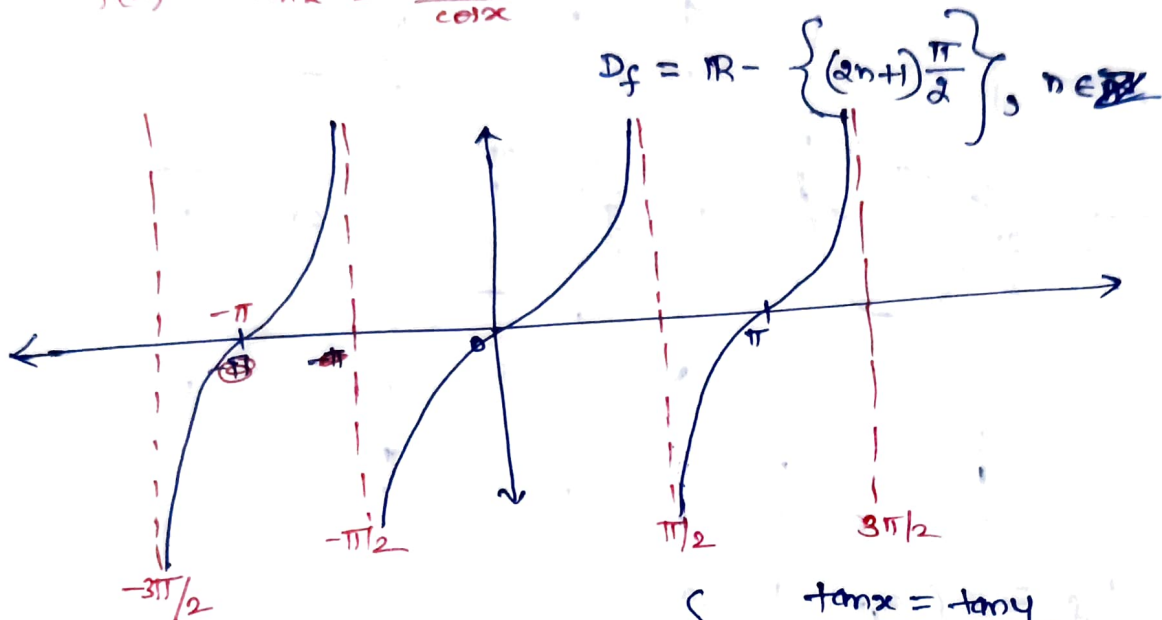
Even function

$$\cos\left((2n+1)\frac{\pi}{2}\right) = 0, n \in \mathbb{Z} ; \cos n\pi = (-1)^n ; n \in \mathbb{Z}$$

$$\cos x = \cos y$$

$$y = 2n\pi \pm x, n \in \mathbb{Z}$$

13. $f(x) = \tan x = \frac{\sin x}{\cos x}$



$R_f = \mathbb{R}$
 Period = π
 odd function
 Not one-one

$$\tan x = \tan y$$

$$y = \pi + x$$

$$2\pi + x$$

$$3\pi + x$$

$$4\pi + x$$

$$y = n\pi + x, n \in \mathbb{Z}$$

14. $f(x) = \cot x = \frac{\cos x}{\sin x}$

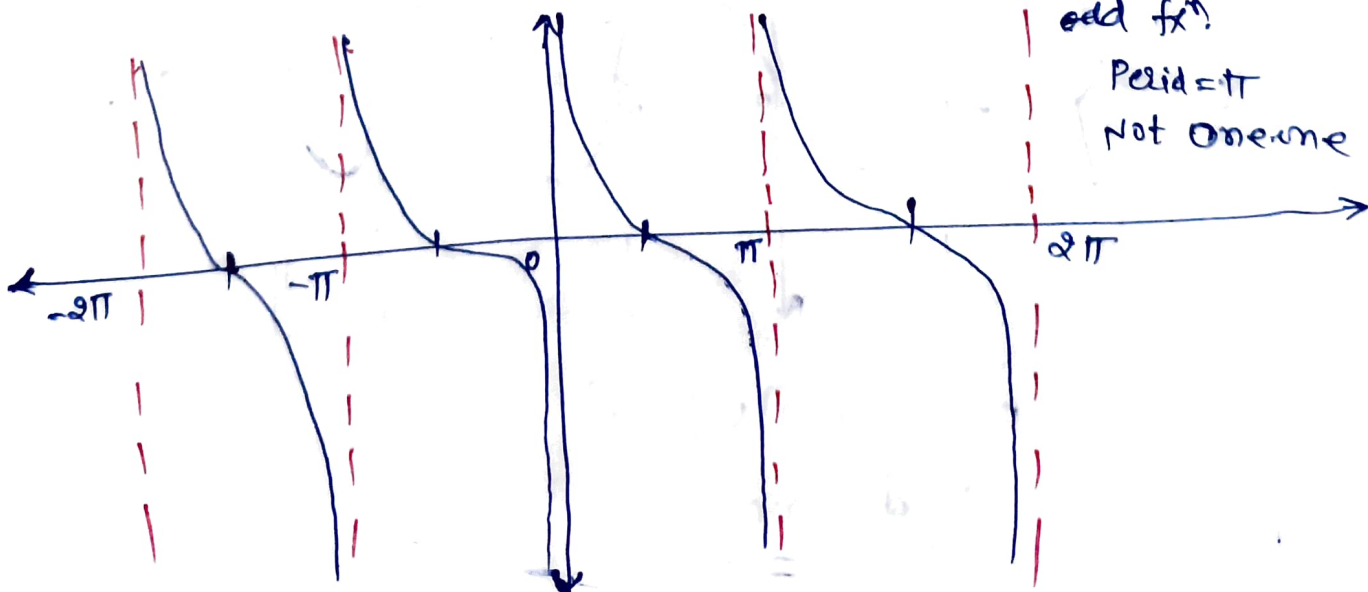
$$D_f = \mathbb{R} - \{n\pi\}; n \in \mathbb{Z}$$

$$R_f = \mathbb{R}$$

odd f(x)

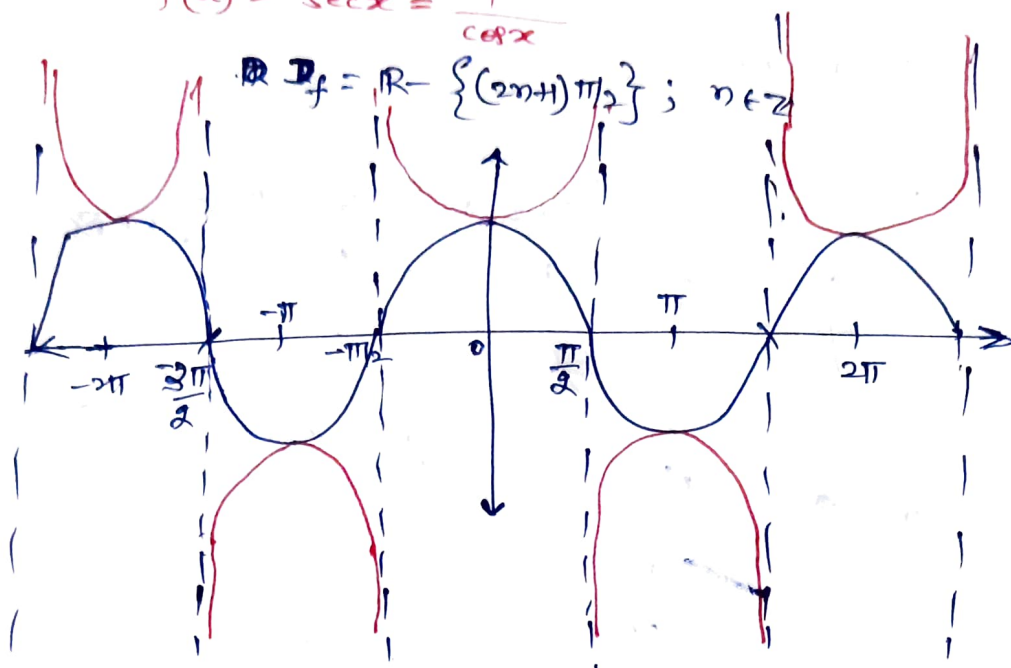
Period = π

Not One-one



15.

$$f(x) = \sec x = \frac{1}{\cos x}$$



$$R_f = (-\infty, -1] \cup [1, \infty)$$

Even function

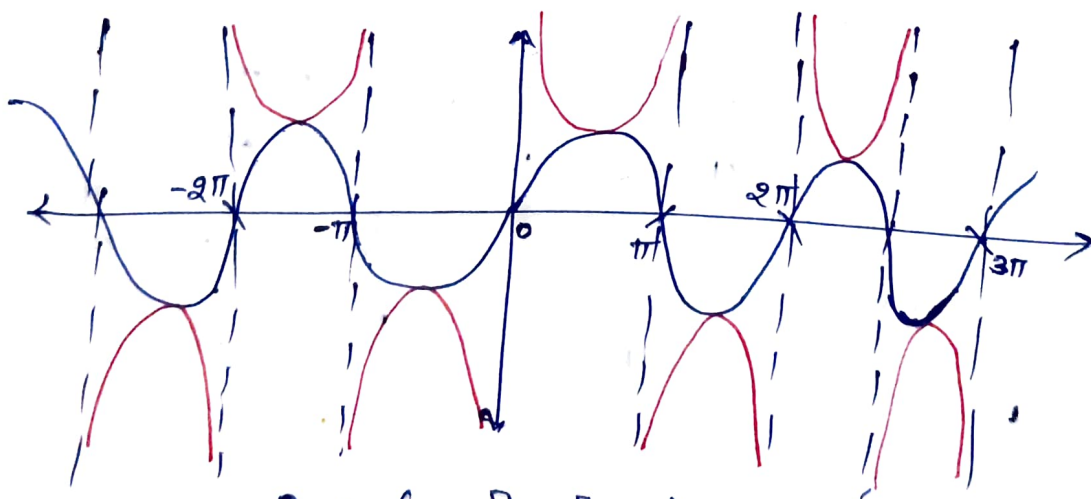
Period = 2π

Not one-one

16.

$$f(x) = \csc x = \frac{1}{\sin x}$$

$$\mathbb{D}_f = \mathbb{R} - \{n\pi\}; n \in \mathbb{Z}$$



$$R_f = (-\infty, -1] \cup [1, \infty)$$

Odd function

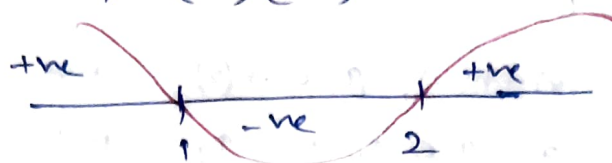
Period = 2π

Not one-one

Sign Method (Wave curve method) :

Solve (i) $f(x) = x^2 - 3x + 2 > 0$

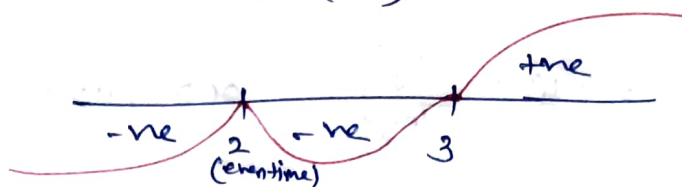
$$\Rightarrow (x-1)(x-2) > 0$$



$$x \in (-\infty, 1) \cup (2, \infty)$$

(ii)

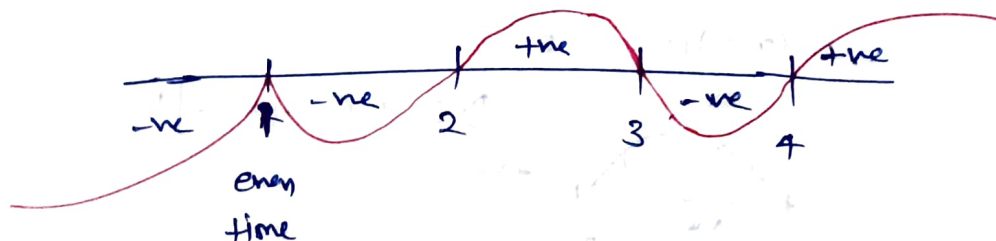
$$f(x) = (x-2)^2 (x-3) > 0$$



$$x \in (3, \infty)$$

(iii)

$$f(x) = \frac{(x-1)^2 (x-2)}{(x-3)(x-4)} \geq 0$$

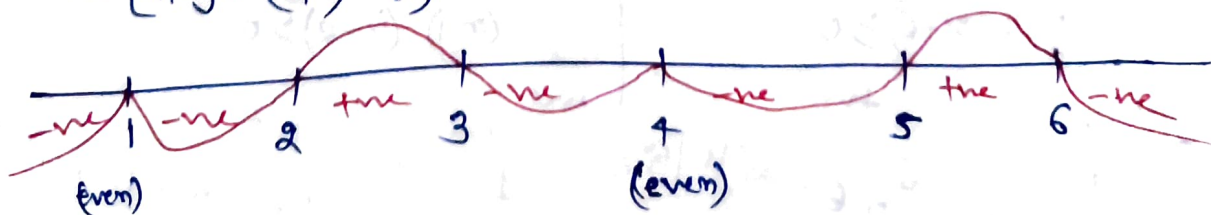


$$x \in [2, 3) \cup (4, \infty) \cup \{1\}$$

(iv)

$$f(x) = \frac{(x-1)^2 (x-2) (x-3)^3}{(x-4)^4 (x-5) (6-x)} \geq 0$$

$$x \in [2, 3] \cup (5, 6) \cup \{1\}$$



Rule to find domain:

- (i) Expression under even square root $(\sqrt[n]{f(x)}) \geq 0$
(i.e. $f(x) \geq 0$)
- (ii) Denominator $\neq 0$
- (iii) If $f(x) = g(x) \pm h(x)$, $g(x) \cdot h(x)$ and domain of $g(x)$ and $h(x)$ are D_1 and D_2 respectively. Then domain of $f(x)$ is $(D_1 \cap D_2)$
- (iv) If $f(x) = \frac{g(x)}{h(x)}$ Then $D_f = D_1 \cap D_2 - \{x \mid h(x) = 0\}$
- (v) If $f(x) = \log_a x = \frac{\log_e x}{\log_e a}$; $x > 0, a > 0, a \neq 1$

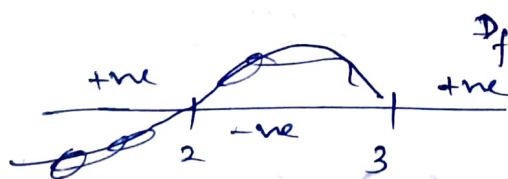
Que: Find Domain

(i) $f(x) = \sqrt{x^2 - 5x + 6}$

For $f(x)$ to be defined

$$x^2 - 5x + 6 \geq 0$$

$$(x-3)(x-2) \geq 0$$



$$D_f = x \in (-\infty, 2] \cup [3, \infty)$$

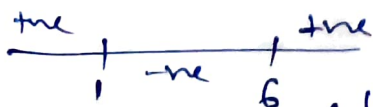
(ii) $f(x) = \sqrt{x^2 - 7x + 6} + \frac{1}{\sqrt{x^2 - 3x + 2}}$

$$x^2 - 7x + 6 \geq 0$$

$$x^2 - 6x - x + 6 \geq 0$$

$$6x(x-6) - 1(x-6) \geq 0$$

$$(x-1)(x-6) \geq 0$$



$$D_1 = x \in (-\infty, 1] \cup [6, \infty)$$

$$x^2 - 3x + 2 > 0$$

$$x^2 - 2x - x + 2 > 0$$

$$x(x-2) - 1(x-2) > 0$$

$$(x-1)(x-2) > 0$$



$$D_2 = x \in (-\infty, 1) \cup (2, \infty)$$

$$D_f = D_1 \cap D_2$$

$$D_f = (-\infty, 1) \cup [6, \infty)$$

$$(iii) \quad f(x) = \sqrt{x^2 + 7x + 12} + \frac{1}{\sqrt{x^2 + 5x - 6}}$$

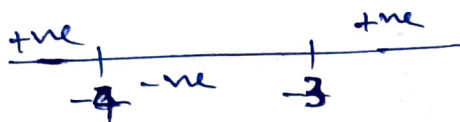
D_1

$$x^2 + 7x + 12 > 0$$

$$x^2 + 4x + 3x + 12 > 0$$

$$x(x+4) + 3(x+4) > 0$$

$$(x+3)(x+4) > 0$$



$$D_1: x \in (-\infty, -4] \cup [-3, \infty)$$

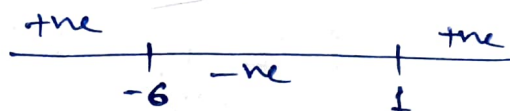
D_2

$$x^2 + 5x - 6 > 0$$

$$x^2 + 6x - x - 6 > 0$$

$$x(x+6) - (x+6) > 0$$

$$(x+1)(x+6) > 0$$



$$D_2: x \in (-\infty, -6) \cup (-1, \infty)$$

$$D_f = D_1 \cap D_2$$

$$D_f = (-\infty, -6) \cup (-1, \infty)$$

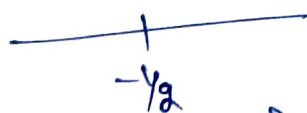
$$(iv) \quad f(x) = \sqrt{\frac{2x+1}{x^3 - 3x^2 + 2x}}$$

Wrong method $2x+1 > 0$

$$2x > -1$$

$$x > -\frac{1}{2}$$

$$x > -\frac{1}{2}$$



$$D_1: x \in [-\frac{1}{2}, \infty)$$

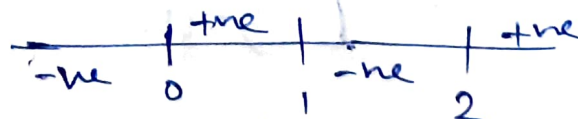
$$x^3 - 3x^2 + 2x > 0$$

$$x(x^2 - 3x + 2) > 0$$

$$x(x^2 - 2x - x + 2) > 0$$

$$x[x(x-2) - (x-2)] > 0$$

$$x[(x-1)(x-2)] > 0$$



$$D_2: (0, 1) \cup (2, \infty)$$

$$D_f = D_1 \cap D_2 = \{x | x^3 - 3x^2 + 2x = 0\}$$

$$D_f = D_1 \cap D_2 - \{0, 1, 2\}$$

$$D_f = (0, 1) \cup (2, \infty) - \{0, 1, 2\}$$

$$D_f = (0, 1) \cup (2, \infty)$$

Right Method:

for $f(x)$ to be defined

$$\frac{2x+1}{x(x-1)(x-2)} \geq 0$$

$$\begin{array}{ccccccc} & +ve & & -ve & & +ve & & -ve & & +ve \\ & | & & | & & | & & | & & | \\ -1/2 & & 0 & & 1 & & 2 & & \end{array}$$

$$x \in (-\infty, -1/2] \cup (0, 1) \cup (2, \infty)$$

Que:

$$f(x) = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$$

$$\therefore \log_{10}(1-x) = \frac{\log_e 10}{\log_e(1-x)} ; \text{ where } 10 > 0, 1-x > 0 \text{ and } 1-x \neq 1 \Rightarrow x \neq 0$$

$$1 > x \Rightarrow x < 1$$

$$D_1: (-\infty, 0) \cup (0, 1)$$

D₂

$$\sqrt{x+2} \quad x+2 \geq 0$$

$$x \geq -2$$

$$D_2: [-2, \infty)$$

$$D_f = D_1 \cap D_2$$

$$= [-2, 1) - \{0\}$$

Que:

$$f(x) = \frac{2}{(x^2-4)} + \log_{10}(x^3-x)$$

For $f(x)$ to be defined

$$x^2-4 \neq 0$$

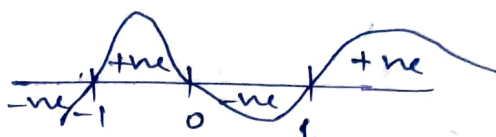
$$x^2 \neq 4$$

$$x \neq \pm 2$$

$$D_1: x \in (-\infty, -2) \cup (-2, 2) \cup (2, \infty)$$

$$x^3-x > 0$$

$$x(x-1)(x+1) > 0$$



$$x \in (-1, 0) \cup (1, \infty)$$

$$D = D_1 \cap D_2$$

$$D = (-1, 0) \cup (1, 2) \cup (2, \infty)$$

Que:

$$f(x) = \log_{x+1}(x^2-3x+2)$$

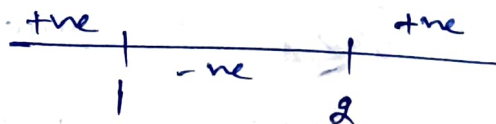
To be defined
f(x)

$$x+1 > 0 ; x+1 \neq 1 ; x^2-3x+2 > 0$$

$$x > -1 ; x \neq 0 ; x^2-2x-x+2 > 0$$

$$x(x-2)-1(x-2) > 0$$

$$(x-1)(x-2) > 0$$



$$x \in (-\infty, 1) \cup (2, \infty)$$

$$D_f = (-1, 0) \cup (0, 1) \cup (2, \infty)$$

Que:

$$f(x) = \sqrt{\log_e \left(\frac{5x-x^2}{4} \right)}$$

$$\log_e \left(\frac{5x-x^2}{4} \right) \geq 0 \quad \text{and} \quad \frac{5x-x^2}{4} > 0$$

$$\frac{5x-x^2}{4} \geq 1$$

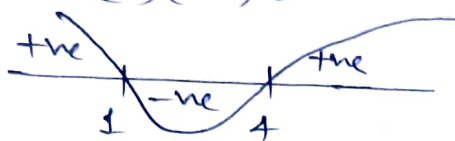
$$\text{and} \quad x(5-x) > 0$$

$$\Rightarrow x^2-5x+4 \leq 0$$

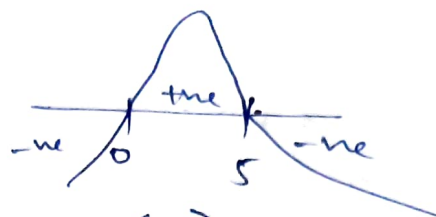
$$x^2 - 4x - x + 4 \leq 0$$

$$x(x-4) - 1(x-4) \leq 0$$

$$(x-1)(x-4) \leq 0$$



$$D_1: x \in [1, 4]$$



$$D_2: x \in (0, 5)$$

$$D_f = D_1 \cap D_2$$

$$D_f = [1, 4]$$

Que:

$$f(x) = \sqrt{\log_{1/2} \left(\frac{5x-x^2}{4} \right)}$$

for $f(x)$ to be defined

$$\log_{1/2} \left(\frac{5x-x^2}{4} \right) \geq 0 \quad \text{and} \quad \left(\frac{5x-x^2}{4} \right) > 0$$

$$\left(\frac{1}{2} \right)^{\log_{1/2} \left(\frac{5x-x^2}{4} \right)} \leq \left(\frac{1}{2} \right)^0 \Rightarrow \frac{5x-x^2}{4} \leq 1$$

$$5x-x^2 \leq 4$$

$$\Rightarrow x^2 - 5x + 4 \geq 0$$

$$\Rightarrow x \in (-\infty, 1] \cup [4, \infty)$$

$$\Rightarrow x \in (-\infty, 1] \cup [4, \infty)$$

$$\text{and } 5x-x^2 > 0$$

$$\text{and } x(5-x) > 0$$

$$\text{and } x(5-x) > 0$$

$$\text{and } x \in (0, 5)$$

$$\text{and } x \in (0, 5)$$

$$D_f = D_1 \cap D_2$$

$$= (0, 1] \cup [4, 5)$$

In q^x

when $2 < 3$

$$\text{but } \left(\frac{1}{2} \right)^2 > \left(\frac{1}{2} \right)^3$$

Que:

$$f(x) = \log_{10} \log_{10} \log_{10} x$$

$$x > 0 \quad \left| \begin{array}{l} \log_{10} x > 0 \\ (10)^{\log_{10} x} > (10)^0 \\ x > 1 \end{array} \right. \quad \left| \begin{array}{l} \log_{10} \log_{10} x > 0 \\ 10^{(\log_{10} \log_{10} x)} > (10)^0 \\ \log_{10} x > 1 \\ 10^{(\log_{10} x)} > (10)^1 \\ x > 10^1 \end{array} \right.$$

$$x > 10^0$$

$$D_f = (10^1, \infty)$$

Que:

$$f(x) = \log_4 \log_2 \log_{\frac{1}{2}} x$$

$$x > 0 \quad \left| \begin{array}{l} \log_{\frac{1}{2}}(x) > 0 \\ \left(\frac{1}{2}\right)^{\log_{\frac{1}{2}}(x)} < \left(\frac{1}{2}\right)^0 \\ x < 1 \end{array} \right. \quad \left| \begin{array}{l} \log_2 \log_{\frac{1}{2}} x > 0 \\ 2^{(\log_2 \log_{\frac{1}{2}} x)} > (2)^0 \\ \log_{\frac{1}{2}} x > 1 \\ x < \frac{1}{2} \end{array} \right.$$

$$D_f = (0, \frac{1}{2})$$

Que:

$$f(x) = \log_{100} x$$

$$\left(\frac{2 \log_{10} x + 1}{-x} \right)$$

$$100x > 0 \quad \left| \begin{array}{l} 100x \neq 1 \\ x \neq \frac{1}{100} \end{array} \right. \quad \left| \begin{array}{l} \frac{2 \log_{10} x + 1}{-x} > 0 \\ \log_{10} x > -\frac{1}{2} \\ x > \frac{1}{\sqrt{10}} \end{array} \right.$$

$$x \in (0, \infty) - \left\{ \frac{1}{100} \right\}$$

$$\frac{2 \log_{10} x + 1}{-x} > 0$$

← Doubt ??
solve by $g(x)$
 $h(x)$

$$\log_{10} x > -\frac{1}{2}$$

$$x > \frac{1}{\sqrt{10}}$$

$$x \in (0, \frac{1}{\sqrt{10}}) \cup (\frac{1}{\sqrt{10}}, \infty)$$

$$x \in (0, \frac{1}{\sqrt{10}})$$

$$\begin{array}{c} + \quad + \quad + \\ \hline 0 \quad \frac{1}{100} \quad \frac{1}{\sqrt{10}} \\ \hline 0 \quad 0 \quad 0 \\ \hline 0 \quad 0 \quad 0 \end{array}$$

$$D_f = x \in (0, \frac{1}{\sqrt{10}}) - \left\{ \frac{1}{100} \right\}$$

** Rule to find for Range :-

(i) If Domain = ~~$\mathbb{R}$~~ or ~~$\mathbb{R}^n$~~ {finite points} Then find value of y in these points we will find Range.

eg: (i) $f: A \rightarrow \mathbb{R}$ s.t. $f(x) = x^2$
and $A = \{1, 2, 3\}$

$$R_f = \{f(1), f(2), f(3)\}$$

$$R_f = \{1, 4, 9\}$$

(ii)

$$f(x) = 1 - x \quad \text{for } x \in \mathbb{R}$$

$$n_R = \text{maximum}$$

$$D_f = \{3, 4, 5\}$$

$$R_f = \{4, 3, 2\}$$

(ii)

If Domain = $\mathbb{R}$ or $\mathbb{R}^n$ {finite points} Then find x in terms of y and find those value of y for which x is defined.

eg: (i) $y = f(x) = \frac{x}{x+1}$

$$D_f = \mathbb{R} - \{-1\}$$

$$\text{Now, } y = \frac{x}{x+1}$$

$$\Rightarrow xy + y = x$$

$$\Rightarrow x(1-y) = y$$

$$\Rightarrow x = \frac{y}{1-y}$$

$$\boxed{\text{Range} \neq y \in \mathbb{R} - \{1\}}$$

(b) $f(x) = y = \frac{x^2}{x^2 + 1}$

$\mathcal{D}_f = \mathbb{R}$

$x^2 y + y = x^2 \Rightarrow x^2 y - x^2 = -y$

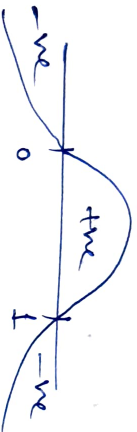
$\Rightarrow x^2(y-1) = -y$

$\Rightarrow x^2 = \frac{y}{1-y}$

$\Rightarrow x = \pm \sqrt{\frac{y}{1-y}}$

For x to be defined

$\frac{y}{1-y} \geq 0$



Range $\in [0, 1]$

(c)

$y = \frac{x - [x]}{1 + \{x\}}$

$y = \frac{\{x\}}{1 + \{x\}}$

Domain: $x \in \mathbb{R}$

$\Rightarrow \{x\}y + y = \{x\}$

$\Rightarrow \{x\}(y-1) = -y$

$\Rightarrow \{x\} = \frac{y}{1-y}$

$\because 0 \leq \{x\} < 1$

$\Rightarrow 0 \leq \frac{y}{1-y} < 1$

$\Rightarrow \frac{y}{1-y} \geq 0 \text{ and } \frac{y}{1-y} < 1$

$$\frac{y}{1-y} > 0$$

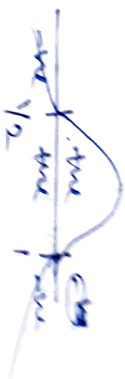
and

$$\frac{y}{1-y} - 1 < 0$$



and

$$\frac{2y-1}{1-y} < 0$$



$$[0, 1]$$

and

$$(-\infty, 1/2) \cup (1, \infty)$$

$$\text{So Range} = y = [0, 1]$$

$$[0, 1/2]$$

Find the Range

$$\textcircled{1} f(x) = y = \frac{x}{x^2 + 1}$$

$$D_f = \mathbb{R}$$

$$x^2 y + y = x$$

$$\Rightarrow x^2 y - x + y = 0$$

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

If $y \neq 0$

$$D > 0 \quad \left(\begin{smallmatrix} x \\ \text{to be real} \end{smallmatrix} \right)$$

$$1 - 4 \cdot y \cdot y > 0$$

$$\Rightarrow (1 - 2y)(1 + 2y) > 0$$



$$y \in [1/8, 1/2]$$

Composite Function $\Rightarrow$

Let $f: A \rightarrow B$ and $g: C \rightarrow D$ s.t.

$f(A) \subseteq C$, Then We define composition of g and f is denoted by $g \circ f: A \rightarrow D$ and is defined as

$$g \circ f(x) = g(f(x))$$

e.g.

$$f: \mathbb{R} \rightarrow \mathbb{R} ; g: \mathbb{R} \rightarrow \mathbb{R} \text{ s.t.}$$

$$f(x) = \sin x ; g(x) = x^2$$

$$f \circ g(x) = f(g(x))$$

$$= f(x^2)$$

$$\Rightarrow f \circ g(x) = \sin x^2$$

$$\text{and } g \circ f(x) = g(f(x))$$

$$= g(\sin x)$$

$$= \sin^2 x$$

Que:

$$\text{If } f(x) = \frac{1}{1-x}, x \neq 1$$

$$\text{Then } f \circ f \circ f(x) = x \quad \forall x \in \mathbb{R} - \{1\}$$

$$f \circ f \circ f(x) = f \circ f(f(x)) ; x \neq 1$$

$$= f \circ f\left(\frac{1}{1-x}\right) ; x \neq 1$$

$$= f\left(f\left(\frac{1}{1-x}\right)\right)$$

$$= f\left[\frac{1}{1-\frac{1}{1-x}}\right] = f\left[\frac{x-1}{x}\right] ; x \neq 0$$

$$= \frac{1}{1-\frac{1}{1-\frac{1}{1-x}}} = \frac{x}{1} = x$$

$\therefore f \circ f \circ f(x) = x \quad \forall x \in \mathbb{R} - \{1\}$ [here f is also not defined at $x=0$]

Que:

$$f(x) = \begin{cases} x+1 & ; x \leq 1 \\ 2x+1 & ; 1 < x \leq 2 \end{cases}$$

$$\text{and } g(x) = \begin{cases} x^2 & ; -1 \leq x < 2 \\ x+2 & ; 2 \leq x \leq 3 \end{cases}$$

find (i) $f \circ g(x) = f(g(x)) = \begin{cases} f(x^2) & ; -1 \leq x < 2 \\ f(x+2) & ; 2 \leq x \leq 3 \end{cases}$

$$f(g(x)) = \begin{cases} x^2+1 & ; -1 \leq x < 2, x^2 \leq 1 \\ 2x^2+1 & ; 2 \leq x < 3, 2 \leq x^2 \leq 2 \\ x+2+1 & ; 2 \leq x \leq 3, x+2 \leq 1 \\ 2(x+2)+1 & ; 2 \leq x \leq 3, 1 < x+2 \leq 2 \end{cases}$$

$$f(g(x)) = \begin{cases} x^2+1, & -1 \leq x \leq 1 \\ 2x^2+1, & 1 < x \leq \sqrt{2} \\ \text{---} & \end{cases}$$

$$(ii) g \circ f(x) = g(f(x)) = \begin{cases} g(x+1) & ; x \leq 1 \\ g(2x+1) & ; 1 < x \leq 2 \end{cases}$$

$$= \begin{cases} (x+1)^2 & ; x \leq 1, -1 \leq x+1 < 2 \\ x+1+2 & ; x \leq 1, 2 \leq x+1 \leq 3 \\ (2x+1)^2 & ; 1 < x \leq 2, -1 \leq 2x+1 < 2 \\ 2x+1+2 & ; 1 < x \leq 2, 2 \leq 2x+1 \leq 3 \end{cases}$$

$$g \circ f(x) = \begin{cases} (x+1)^2, & -2 \leq x \leq 1 \\ (2x+1)^2, & -1 \leq x < 2 \\ 2x+3, & \end{cases}$$

$$(iii) f \circ f(x) = f(f(x)) = \begin{cases} f(x+1), & x \leq 1 \\ f(2x+1), & 1 < x \leq 2 \end{cases}$$

$$= \begin{cases} x+2, & x \leq 1, x+1 \leq 1 \\ 2(x+1)+1, & x \leq 1, 1 < x+1 \leq 2 \\ 2x+2, & 1 < x \leq 2, 2x+1 \leq 1 \\ 2(2x+1)+1, & 1 < x \leq 2, 1 < 2x+1 \leq 2 \end{cases}$$

$$= \begin{cases} x+2, & x \leq 0 \\ 2x+3, & 0 < x \leq 1 \end{cases}$$

$$(iv) g \circ g(x) = g(g(x)) = \begin{cases} g(x^2); & -1 \leq x < 2 \\ g(x+2); & 2 \leq x \leq 3 \end{cases}$$

$$= \begin{cases} x^4; & -1 \leq x < 2, -1 \leq x^2 < 2 \\ x^2+2; & -1 \leq x < 2, 2 \leq x^2 \leq 3 \\ (x+2)^2; & 2 \leq x \leq 3, -1 \leq x+2 < 2 \\ (x+2)+2; & 2 \leq x \leq 3, 2 \leq x+2 \leq 3 \end{cases}$$

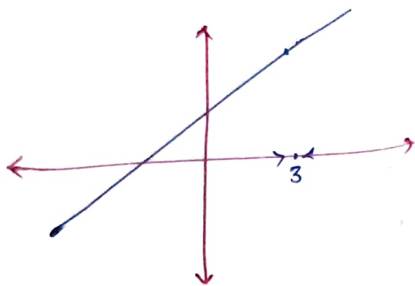
LIMITS

##

Introduction with limits:

Let
 $f(x) = x + 3$

Clearly, $f(x)$ is defined
 $\forall x \in \mathbb{R}$

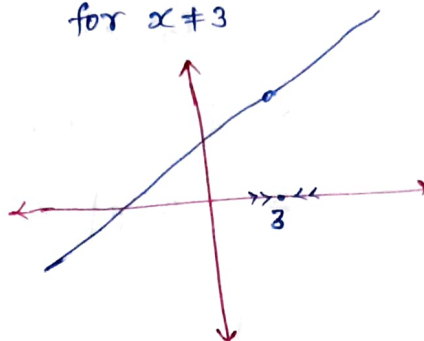


$$x \rightarrow 3 \Rightarrow f(x) \rightarrow 6$$

;

$$g(x) = \frac{x^2 - 9}{x - 3}$$

But $g(x)$ is not defined
 for $x \neq 3$



$$x \rightarrow 3 \Rightarrow g(x) \rightarrow 6$$

Limits: - Defⁿ: (1) A function $f(x)$ has limit at point $x = a$
 i.e. $\lim_{x \rightarrow a} f(x)$ exists if

- (i) $\lim_{x \rightarrow a^+} f(x)$ exists
- (ii) $\lim_{x \rightarrow a^-} f(x)$ exists
- (iii) $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$

Defⁿ: (2) A number $L \in \mathbb{R}$ is said to be limit of $f(x)$ at
 $x = a$ if for given $\epsilon > 0 \rightarrow \exists \delta > 0$ s.t.

$$|f(x) - L| < \epsilon \text{ Whenever } 0 < |x - a| < \delta \cap D_f$$

i.e. When

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$$

$$\Rightarrow -\delta < x - a < \delta \Rightarrow -\epsilon < f(x) - L < \epsilon$$

$$\Rightarrow a - \delta < x < a + \delta \Rightarrow L - \epsilon < f(x) < L + \epsilon$$

$$\Rightarrow x \in (a - \delta, a + \delta) - \{a\} \cap D_f \Rightarrow f(x) \in (L - \epsilon, L + \epsilon)$$

e.g.:

$$f(x) = x^2$$

$$\text{at } x = 2$$

$$\text{as } x \rightarrow 2 \Rightarrow f(x) \rightarrow 4$$

$$|f(x) - 4| < \epsilon$$

$$\Rightarrow |x^2 - 4| < \epsilon$$

$$\Rightarrow |x - 2| |x + 2| < \epsilon$$

$$\Rightarrow |x - 2| < \frac{\epsilon}{|x + 2|} = \delta$$

Que: $\rightarrow$ (i)

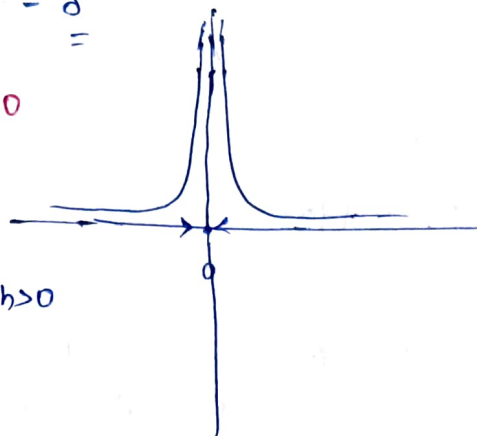
$$f(x) = \frac{1}{x^2} \text{ at } x = 0$$

LHL $\lim_{x \rightarrow 0^-} f(x)$

$$\text{Put } x = 0 - h, h \rightarrow 0^+, h > 0$$

$$\lim_{h \rightarrow 0} f(0 - h) = \lim_{h \rightarrow 0} f(-h)$$

$$= \frac{1}{(-h)^2} = \frac{1}{h^2} = \infty$$



RHL

$$\lim_{x \rightarrow 0^+} f(x)$$

$$\text{Put } x = 0 + h, h \rightarrow 0^+, h > 0$$

$$\lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} \frac{1}{h^2} = \infty$$

$$\therefore \boxed{\lim_{x \rightarrow 0} f(x) = \infty} \text{ DNE}$$

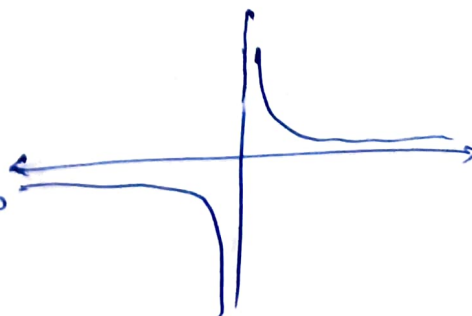
Que: (ii) $f(x) = \frac{1}{x}$ at $x = 0$

LHL

$$\lim_{x \rightarrow 0^-} f(x)$$

$$\text{Put } x = 0 - h, h \rightarrow 0^+, h > 0$$

$$\lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} \frac{1}{-h} = -\infty$$



RHL

$$\lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x} = \infty$$

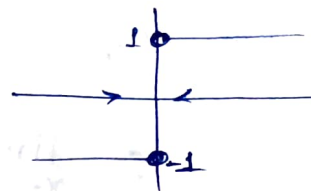
$$\therefore \lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$$

So, limit DNE

(iii)

$$f(x) = \frac{|x|}{x} \text{ at } x=0$$

$$f(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$



LHL

$$\lim_{x \rightarrow 0^-} f(x)$$

Put $x = 0-h$, $h \rightarrow 0$, $h > 0$

$$\lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} \frac{|-h|}{-h} = \lim_{h \rightarrow 0} \frac{h}{-h} = -1$$

RHL

$$\lim_{x \rightarrow 0^+} f(x)$$

$$\lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} \frac{|h|}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

$$\therefore \text{LHL} \neq \text{RHL}$$

So, limit DNE

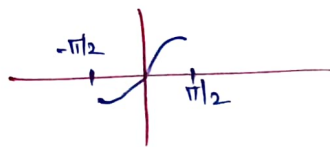
(iv) $f(x) = [\sin x]$ at $x=0$

LHL

$$\lim_{x \rightarrow 0^-} f(x)$$

Put $x = 0-h$, $h \rightarrow 0$, $h > 0$

$$\lim_{h \rightarrow 0} [\sin(-h)] = \lim_{h \rightarrow 0} [-\sin h] = -1$$



RHL

$$\lim_{x \rightarrow 0^+} f(x)$$

$$\lim_{h \rightarrow 0} [\sin h] = 0$$

$$\Rightarrow \text{LHL} \neq \text{RHL}$$

$\Rightarrow$ limit DNE

V) $\lim_{x \rightarrow 0} [\cos x]$

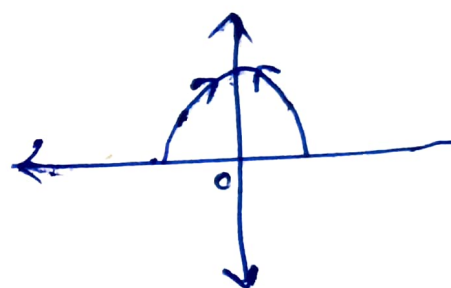
LHL $\lim_{x \rightarrow 0^-} f(x)$

$$\lim_{h \rightarrow 0} [\cos(-h)] = \lim_{h \rightarrow 0} [\cos h] = 0$$

HL $\lim_{x \rightarrow 0^+} f(x)$

$$\lim_{h \rightarrow 0} [\cos h] = \lim_{h \rightarrow 0} [\cos h] = 0$$

$$\therefore \lim_{x \rightarrow 0} [\cos x] = 0$$



$$\lim_{x \rightarrow 0^+} \left[\frac{\tan x}{x} \right] = 1$$

Que: (viii)

$$\lim_{x \rightarrow 0^+} \sin \sqrt{x}$$

//
oscillates b/w
-1 and 1

SO limit DNE.



(ix)

$$\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$$

$$\because 180^\circ = \pi \text{ Radian}$$

$$1^\circ = \frac{\pi}{180} \text{ Radian}$$

$$\lim_{x \rightarrow 0} \frac{\sin \frac{\pi}{180} x}{x} = \lim_{x \rightarrow 0} \frac{\sin \frac{\pi}{180} x}{\frac{\pi}{180} x} \times \frac{\pi}{180} = \frac{\pi}{180}$$



Algebra of limits :-

① If $\lim_{x \rightarrow a} f(x) = l_1$ and $\lim_{x \rightarrow a} g(x) = l_2$ Then

$$(i) \lim_{x \rightarrow a} [f(x) \pm g(x)] = l_1 \pm l_2$$

$$(ii) \lim_{x \rightarrow a} f(x) \cdot g(x) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) \\ = l_1 \cdot l_2$$

$$(iv) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{l_1}{l_2} \quad \left(\text{Provided } g(a) \neq 0 \right)$$

② If $\lim_{x \rightarrow a} f(x) = l_1$ exists finitely and g is continuous at $x = a$ Then

$$\lim_{x \rightarrow a} g[f(x)] = g(l_1)$$

$$\text{i.e. } \lim_{x \rightarrow a} (g \circ f)(x) = \lim_{x \rightarrow a} g[f(x)] \\ = g\left[\lim_{x \rightarrow a} f(x)\right]$$

e.g.

$$g(x) = [x], \quad f(x) = \sin x \\ \text{At } x = 0$$

$$g \circ f(x) = g(f(x)) = g(\sin x) = [\sin x]$$

$$\lim_{x \rightarrow 0} [\sin x] \neq \left[\lim_{x \rightarrow 0} \sin x \right] \quad \left[\because g(x) \text{ is not continuous at } x = 0 \right]$$

LHS

$$\lim_{x \rightarrow 0} [\sin x]$$

$$\text{LHL} = 0, \quad \text{RHL} = -1 \\ \text{DNE}$$

RHS

$$\left[\lim_{x \rightarrow 0} \sin x \right] = 0$$

$$\text{eg: } g(x) = |x|; \quad f(x) = \sin x$$

$$g \circ f(x) = g(f(x)) = g(\sin x) = |\sin x|$$

$$\therefore \lim_{x \rightarrow 0} |\sin x| = \left| \lim_{x \rightarrow 0} \sin x \right|$$

$\because g(x)$ is a cts fn.

③ If $f(x) > 0$ or $f(x) \geq 0$ in neighbourhood of $x=a$ and $\lim_{x \rightarrow a} f(x)$ exists.

Then $\lim_{x \rightarrow a} f(x) \geq 0$

e.g.

$$f(x) = \frac{1}{x} < \frac{1}{x} >$$

$$= \left\{ \frac{1}{2}, \frac{1}{3}, \dots \right\}$$

$$\frac{\tan x}{x} > \frac{\sin x}{x}$$

in nbd of $x=0$

$$\text{but } \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\& \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\left\{ f(x) = \frac{\tan x}{x} - 1 \right\}$$

4. Similarly if $f(x) < 0$ or $f(x) \leq 0$ in nbd of $x=a$ and $\lim_{x \rightarrow a} f(x)$ exists

Then

$$\lim_{x \rightarrow a} f(x) \leq 0$$

5. If $f(x) > g(x)$ or $f(x) \geq g(x)$ in the nbd of $x=a$ and $\lim_{x \rightarrow a} f(x)$ & $\lim_{x \rightarrow a} g(x)$ exists.

$$\therefore \lim_{x \rightarrow a} f(x) \geq \lim_{x \rightarrow a} g(x)$$

Que: - (i) $\lim_{x \rightarrow 0} \sin|x|$

LHL $\lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} \sin|-h| = \lim_{h \rightarrow 0} \sin h = 0$

RHL $\lim_{h \rightarrow 0} f(h) = \lim_{h \rightarrow 0} \sin h = 0$

$$\Rightarrow \lim_{x \rightarrow 0} \sin|x| = 0$$

$$\sin|x| = \begin{cases} \sin x ; x \geq 0 \\ -\sin x ; x < 0 \end{cases}$$

(ii) $\lim_{x \rightarrow 0} |\sin x|$

$$\therefore |\sin x| = \begin{cases} \sin x ; x > 0 \\ -\sin x ; x < 0 \end{cases}$$